Overview

7 Variance Estimation

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7 Variance Estimation

- The variance estimators introduced in Chapters 2–6 were derived for estimators of population totals and means, but not directly apply for estimators of **nonlinear** population characteristics.

- An important **class** of nonlinear characteristics can be written as a **function** of a vector of population totals,

$$\mathbf{t} = (t_1, \ldots, t_q)^T, \qquad q \geq 1.$$

- The best-known member of this class is the **population ratio** of two totals, $R_U = g(\mathbf{t})$, with $\mathbf{t} = (t_y, t_x)^T$, given by (provided that $t_x \neq 0$)

$$R_U(\mathbf{t}) = \frac{t_y}{t_x} = \frac{\bar{y}_U}{\bar{x}_U}.$$

- For the remainder of this section, assume an SRS of size n.
- The estimator

$$\widehat{R} = \frac{\widehat{t}_y}{\widehat{t}_x}$$

is commonly referred to as the **ratio estimator**. It is generally **biased** for R_U.

- The **absolute bias relative to the standard error** is bounded above by (Hartley and Ross, 1954)

$$\frac{|\text{bias}(\widehat{R})|}{\text{SE}(\widehat{R})} \leq \sqrt{\frac{1}{n} - \frac{1}{N}}\text{CV}_x,$$

where $\text{CV}_x = \sigma_x / \bar{x}_U$ is the coefficient of variation.

$\Rightarrow$ Thus, the bias is small relative to the estimator's standard error when the sample is sufficiently large and CV_x is not too large.

7.1 Taylor Series Expansion (Linearization)

- We restrict attention to the ratio estimator $\widehat{R}$ under SRS. The same idea applies to other nonlinear characteristics that are differentiable functions of population totals.

- Write
$$\widehat{R} = g(\bar{y}, \bar{x}), \qquad g(a,b) = \frac{a}{b},$$
where $\bar{y}$ and $\bar{x}$ are the sample means.

- A first-order **Taylor expansion** around $(\bar{y}_U, \bar{x}_U)$ gives
$$\widehat{R} = R_U + \frac{1}{\bar{x}_U}(\bar{y} - R_U\,\bar{x}) + \mathrm{o}_p(n^{-1/2}). \tag{7.1}$$

- Let r_n denote the remainder in the Taylor expansion. Under the asymptotic framework introduced in Chapter 2,

$$r_n = o_p(n^{-1/2}) \qquad \text{means that} \qquad \frac{r_n}{n^{-1/2}} \xrightarrow{p} 0.$$

- Ignoring the remainder in (7.1) gives the **linearization**

$$\widehat{R}_0 = R_U + \frac{1}{\bar{x}_U}(\bar{y} - R_U\,\bar{x}).$$

- Define the **linearized variable**

$$z_i = y_i - R_U x_i.$$

Then

$$\widehat{R}_0 - R_U = \frac{\bar{z}}{\bar{x}_U}, \qquad \mathsf{var}(\widehat{R}) \approx \mathsf{var}(\widehat{R}_0) = \frac{1}{\bar{x}_U^2}\mathsf{var}(\bar{z}).$$

- Under SRS, the linearization yields

$$\mathsf{var}(\widehat{R}) \approx \mathsf{var}(\widehat{R}_0) = \left(1 - \frac{n}{N}\right) \frac{\sigma_z^2}{n\,\bar{x}_U^2},$$

  where σ_z^2 is the population variance of the linearized variable.

- Since R_U and $\bar{x}_U$ are unknown, define the **estimated linearized variable**

$$\widehat{z}_i = y_i - \widehat{R} x_i.$$

- A corresponding **variance estimator** is

$$\widehat{\mathsf{var}}(\widehat{R}) = \left(1 - \frac{n}{N}\right) \frac{\widehat{\sigma}_{\widehat{z}}^2}{n\,\bar{x}^2}, \qquad \text{where} \qquad \widehat{\sigma}_{\widehat{z}}^2 = \frac{1}{n-1} \sum_{i \in s} \widehat{z}_i^2.$$

- Under suitable regularity conditions, Taylor linearization can also be applied to **complex sampling designs**; see, e.g., Fuller (2009, Chapter 2.1).
- A main **disadvantage** is that the linearized variable must be derived for each nonlinear estimator separately. This can become cumbersome for complex estimands and sampling designs.
- **Resampling methods** provide an alternative approach to variance estimation that often requires less estimator-specific derivation.

7.2 Resampling Methods

- **Resampling methods** estimate the variance of an estimator by repeatedly constructing **replicate samples** from the observed sample and recomputing the estimator.
- Their main advantage is that they often avoid **estimator-specific analytical derivations**, such as those required for Taylor linearization.
- The price is additional **computational effort**, especially for complex sampling designs.
- We consider two resampling methods:
  a) Bootstrap
  b) Jackknife

Bootstrap

- Let s be an SRS of size n and let $\widehat{\theta}$ be an estimator of the population characteristic θ_U.
- Choose the number of **bootstrap replicates** B.
- For each $b = 1, \ldots, B$:
  1. Draw a **bootstrap sample** s_b^* of size n from s, sampling **with replacement**.
  2. Compute the estimator from s_b^* and denote the result by $\widehat{\theta}_b^*$.
- The resulting values $\widehat{\theta}_1^*, \ldots, \widehat{\theta}_B^*$ approximate the sampling variability of $\widehat{\theta}$.

- The **bootstrap variance estimator** is the sample variance of the replicate estimates:

$$\widehat{\mathrm{var}}_{\mathrm{boot}}\left(\widehat{\theta}\right) = \frac{1}{B-1}\sum_{b=1}^{B}\left(\widehat{\theta}_b^* - \bar{\theta}^*\right)^2, \qquad \bar{\theta}^* = \frac{1}{B}\sum_{b=1}^{B}\widehat{\theta}_b^*.$$

- Under SRS, the ordinary bootstrap is appropriate when the **sampling fraction**

$$f = \frac{n}{N}$$

is small.

- If f is not small, sampling without replacement produces less variability than the ordinary bootstrap. For smooth estimators, this can be accounted for approximately by the **finite population correction** (Wolter, 2007, Chapter 5)

$$(1-f)\,\widehat{\mathrm{var}}_{\mathrm{boot}}(\widehat{\theta}).$$

Bootstrap for Stratified Sampling

- For a stratified design, the bootstrap should **preserve the stratification**.
- For each bootstrap replicate $b = 1, \ldots, B$:
  1. Within each stratum $h = 1, \ldots, H$, draw a bootstrap sample s_{hb}^* of size n_h from s_h, sampling **with replacement**.
  2. Combine the stratum-specific bootstrap samples and compute the replicate estimate $\widehat{\theta}_b^*$.
- If the stratum-specific sampling fractions $f_h = n_h/N_h$ are small, the ordinary within-stratum bootstrap provides a useful approximation.
- If the f_h are not small, a single finite population correction is generally **not sufficient**.

Rao–Wu Rescaling Bootstrap

- If the stratum-specific sampling fractions f_h are not small, the ordinary within-stratum bootstrap does not adequately reflect sampling without replacement.
- The **Rao–Wu rescaling bootstrap** modifies the bootstrap procedure to account for this; see Rao and Wu (1988).
- Each bootstrap replicate is associated with a set of adjusted **replicate weights**. The estimator is recomputed using these weights, yielding replicate estimates

$$\widehat{\theta}_1^*, \ldots, \widehat{\theta}_B^*.$$

- The approach can be applied to many nonlinear estimators without deriving a separate linearized variable, including estimators based on **quantiles**.

Jackknife

- Let s be an SRS of size n from U.
- We are interested in the variance of an estimator $\widehat{\theta}$ of the characteristic θ_U, where θ_U is defined as a **differentiable** function $g(\mathbf{t})$ of population totals $\mathbf{t}$.
- **Differentiability** of g is a *necessary* condition (Krewski and Rao, 1981) for consistency of the (delete-1) jackknife variance estimator.
- In the **delete-1 jackknife**, one observation at a time is omitted from s, and the estimator is recomputed without that observation. We write $\widehat{\theta}_{(-j)}$ for the estimator $\widehat{\theta}$ computed *without* the jth observation ($j \in s$).
- Example: For $\widehat{\theta} := \bar{y}$, $\widehat{\theta}_{(-j)} = 1/(n-1)\sum_{i \in s \setminus j} y_i$.

- The delete-1 jackknife **variance estimator** for $\widehat{\theta}$ is

$$\widehat{\text{var}}_{\text{JK}}(\widehat{\theta}) = \frac{n}{(n-1)} \sum_{j=1}^{n} \left(\widehat{\theta}_{(-j)} - \widehat{\theta} \right)^2 . \tag{7.2}$$

- An **alternative estimator** is obtained by replacing $\widehat{\theta}$ in (7.2) by $\widetilde{\theta}$, where $\widetilde{\theta} = (1/n) \sum_{j=1}^{n} \widehat{\theta}_{(-j)}$; see Wolter (2007, p. 170).

- We denote this alternative estimator by $\widehat{\text{var}}_{\text{JK}^*}$. For nonlinear estimators, typically $\widehat{\theta} \neq \widetilde{\theta}$.

- The variance estimators $\widehat{\text{var}}_{\text{JK}}$ and $\widehat{\text{var}}_{\text{JK}^*}$ are reliable when the sampling fraction f is small.

- If f is **not small**, both variance estimators can be substantially biased. In this case, it is recommended to multiply them by $(1-f)$; see (Wolter, 2007, p. 167).

- The jackknife variance estimator can also be used with **complex designs**; see, e.g., Wolter (2007, Chapter 4).
- The jackknife can also be generalized by deleting $d > 1$ observations at a time, i.e., the **delete-d jackknife**. Shao and Wu (1989) further showed that, with an appropriate choice of d, the delete-d jackknife variance estimator can be consistent for statistics such as the median or a linear combination of quantiles, unlike the delete-1 jackknife.

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